UAV path planning based on improved A * and artificial potential field algorithm

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***Abstract*—The traditional A * algorithm cannot guarantee that the obtained path is optimal in path planning, and the path generally has many inflection points. By improving the weight of the cost function h(n), the optimal path is obtained; Secondly, adding hypothetical target points based on each inflection point as the basis for path partitioning to make the overall path smoother. The traditional APF method has the problems of unreachable targets and local optima. By increasing the target position function and introducing virtual target points, the gravitational and repulsive fields experienced by the drone are changed, enabling the UAV to quickly respond to obstacles and plan the correct path to avoid them. According to the simulation results, the improved fusion of the two algorithms can plan a smooth optimal path and reasonably avoid obstacles.**

Keywords—path planning; A * algorithm; APF; UAV

I. INTRODUCTION

The path planning and obstacle avoidance of drones have always been hot topics in the field of control research. Good path planning algorithms can greatly improve the time and efficiency of drones completing tasks, and effectively reduce unexpected accidents that occur during the task completion process [1]. The main research focus of path planning is to enable unmanned aerial vehicles to automatically avoid obstacles and plan an optimal route to complete tasks in the environment where tasks need to be completed.

In the past, commonly used methods for obstacle avoidance include A * algorithm [2], artificial potential field method [3], genetic algorithm [4], neural network method [5], ant colony algorithm [6], and particle swarm optimization algorithm [7]. Reference [8] improves the number and angle of turns in the path by adding search domains to the A * algorithm and introducing angle search algorithms. Reference [9] proposes a path planning algorithm for material transfer robots by combining A * algorithm with dynamic windows, which effectively reduces the length of the path and makes the planned path smoother. Reference [10] combines A * algorithm with ant colony algorithm to improve the search ability of the algorithm, and uses B-spline curves to fit the path, making it smoother. Reference [11] proposes a combination of APF method, particle swarm optimization algorithm, and genetic algorithm for path planning, which enables the algorithm to have strong optimization ability and convergence speed. Reference [12] changes the distance factor using the artificial potential field method and adopts the method of bypassing obstacles to ensure that the planned path can safely reach the endpoint. Reference [13] adjusts the repulsive coefficient of the artificial potential field method to enable safe obstacle avoidance. Reference [14] adopts the method of planar position sampling, and determines the direction of robot movement by minimizing the weighted sum of gravity and repulsion. From the above improvement methods, it can be seen that they are mainly limited to angle control and changes in gravitational and repulsive coefficients, making the route smoother without considering the timeliness of avoiding obstacles.

The A * algorithm, as a heuristic search algorithm, is usually used for overall path planning. By evaluating the function, a path to the target point can be found in the overall environment. However, due to the uncertainty of the evaluation function, it cannot guarantee that the obtained path is the optimal path, and the search efficiency is not high. The planned path is not smooth and does not conform to the actual flight trajectory of the drone [15]. The artificial potential field method is usually used for local path planning and has good obstacle avoidance ability, which can quickly respond to obstacles. It is computationally simple and easy to manipulate, but traditional artificial potential field methods often suffer from two types of problems: local minima and unreachable targets due to the complexity of obstacles [16]. Therefore, the path planning effect in complex environments is not ideal. This article improves the heuristic function of the A * algorithm and eliminates redundant nodes to calculate the overall path of the UAV. The planned organized path is segmented based on inflection points, and optimized again through APF to achieve good obstacle avoidance effect and shorten the path length again.

II. IMPROVE THE A * ALGORITHM

A. Traditional A * algorithm

The A * algorithm is the most classic heuristic function search algorithm and a relatively universal algorithm. It estimates the value of each point in the environment, compares the value of each point [17], and obtains the optimal path. Its cost function is

$$f(n) = g(n) + h(n) \tag{1}$$

Among them, g(n) represents the cost that has been consumed from the starting point to the current position of the robot in the environment where the target point is located, as shown in Fig 1:

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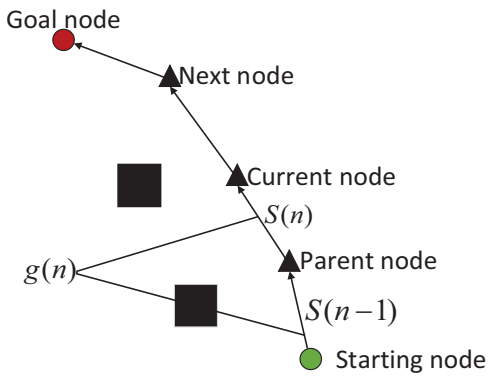

Fig. 1. Relationship between g(n) and nodes

and h(n) represents the cost that needs to be consumed from the current position of the robot to the target point in the environment where the target point is located, which can be expressed as:

$$h(c)=\left|\overrightarrow{NG}\right|=\left|x_G-x_N\right|+\left|y_G-y_N\right| \tag{2}$$

In the formula, $\overrightarrow{NG}$ is the direction vector from the current robot's point to the target point, and the direction vector from the starting point to the target point is defined as $\overrightarrow{SG}$. The angle between the two vectors is set to θ. To enable the algorithm to search more efficiently, we can add vector angle $h(n)$ to heuristic function θ, where θ can be represented as

$$\cos\theta=\frac{\overrightarrow{SG}\bullet\overrightarrow{NG}}{\left|\overrightarrow{SG}\right|\times\left|\overrightarrow{NG}\right|} \tag{3}$$

$$\left|\overrightarrow{SG}\right|=\sqrt{\left(X_G-X_S\right)^2+\left(Y_G-Y_S\right)^2} \tag{4}$$

$$\left|\overrightarrow{NG}\right|=\sqrt{\left(X_G-X_N\right)^2+\left(Y_G-Y_N\right)^2} \tag{5}$$

Since the starting point S and the target point G are fixed, the vector sg remains constant. But for vector $\overrightarrow{SG}$, it changes with the angle θ between the two vectors. Therefore, the smaller the θ value, the fewer expansion nodes, and the more ideal the planned path.

*B. Improve the A * algorithm*

a. Improve h (n)

The traditional A * algorithm can make overall path planning for the target point in the presence of complex obstacles, but due to its simple cost function, the planned path is often not the optimal path, and the search efficiency is not high [18]. Therefore, this article starts with the cost function and improves the traditional A * algorithm to make the path more in line with the actual situation.

As can be seen from the traditional cost function. The main factor affecting the path is $h(n)$. If the actual distance between the robot's current position and the target point is $d(n)$, there will be three situations:

- When $h(n)<d(n)$, the search range is large, resulting in too many search nodes. Although the optimal path can be found, the efficiency is not high;
- When $h(n)=d(n)$, the algorithm can find the shortest distance for the robot to reach the target point;
- When $h(n)>d(n)$ is used, the search range is small, so there are relatively few search nodes and the search speed is fast. However, the obtained path at this moment cannot guarantee whether it is the optimal path, and when $h(n)\gg d(n)$ is used, only $h(n)$ in the cost function can affect the final generation value.

When the drone is located far away from the target point, the weight of $h(n)$ is too small, which will cause too many search nodes and reduce efficiency. Therefore, the weight of the heuristic function value should be increased. When the position of the drone is very close to the target point, the weight of $h(n)$ will become very large, making it difficult to find the optimal path. So make the following modifications to the cost function:

$$f(n)=g(n)+\left(1+\frac{d}{D}\right)h(n) \tag{6}$$

In equation (6), d is the distance between the current position of the robot and the target point, and D is the distance from the starting point to the target point.

b. Eliminate redundant nodes

The path planned by traditional A * algorithm usually has too many inflection points and is not smooth, which leads to the robot being unable to move according to the predetermined trajectory in actual motion [19]. Therefore, it is necessary for us to make the path smoother by reducing redundant inflection points [20]. In this article, based on each turning point, the planned path is divided into several small segments. The intersection point between the middle path and obstacles in the environment is assumed as the target point, and each path is processed separately to obtain a smoother trajectory.

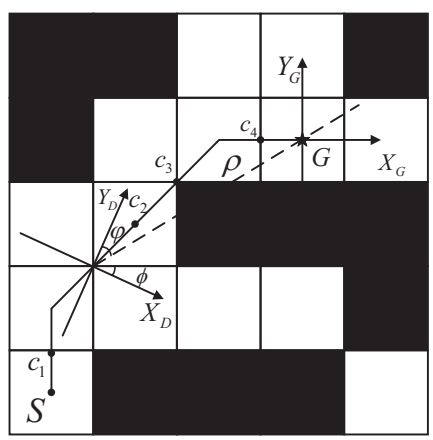

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Fig. 2. Mathematical model for reducing turning points

As shown in Fig 2, S and G are the starting and target nodes on the path, respectively, and the solid line represents the planned path; c_1、c_2、c_3、c_4 is the four sub target points on the change path; ϕ is the angle between the current target coordinate system and the target point coordinate system, and φ is the angle between the current target coordinate system and the target direction vector.

As the robot moves towards the target along this path, all the above variables will change. By inputting this change into the following controller, we can obtain the robot's speed command (v, ω).

$$\begin{cases} v = K_\rho \tanh(K_v \rho) \\ \omega = K_\varphi \varphi + K_\phi \phi \end{cases} \tag{7}$$

To ensure the smoothness of the path, the controller needs to meet the following conditions:

$$\begin{cases} K_v > 0 \\ K_\rho > 0 \\ K_\phi < 0 \\ K_\varphi + K_\phi - K_\rho K_v > 0 \end{cases} \tag{8}$$

Based on the obtained speed command, we can calculate the mathematical model of the robot's node pose $(x、y、\theta)$ at this moment：

$$\begin{cases} x(t+\Delta t) = x(t) - \dfrac{v}{\omega}\sin\theta + \dfrac{v}{\omega}\sin(\theta + \omega\Delta t) \\ x(t+\Delta t) = y(t) - \dfrac{v}{\omega}\cos\theta + \dfrac{v}{\omega}\cos(\theta + \omega\Delta t) \\ \theta(t+\Delta t) = \theta(t) + \omega\Delta t \end{cases} \tag{9}$$

III. Improving the Artificial Potential Field Algorithm

A. Traditional artificial potential field method

The artificial potential field method calculates the repulsive force of obstacles on the target and the gravitational force of the target point on the target point. Under the combined action of these two forces, the target is driven to move towards the target point [21]. The gravitational potential field and repulsive potential field are shown as follows:

$$U_{at}(X) = \frac{1}{2}K\left(X - X_g\right)^2 \tag{10}$$

$$U_{re}(X) = \begin{cases} \dfrac{1}{2}\eta\left[\dfrac{1}{\lambda} - \dfrac{1}{\lambda_0}\right]^2 & \lambda(X, X_{ob}) \le \lambda_0 \\ 0 & \lambda(X, X_{ob}) > \lambda_0 \end{cases} \tag{11}$$

The force acting on the target is the negative gradient at the potential field where the target is located:

$$F_{at}(X) = -K\left(X - X_g\right) \tag{12}$$

$$F_{re}(X) = \begin{cases} \eta\left(\dfrac{1}{\lambda} - \dfrac{1}{\lambda_0}\right)\dfrac{1}{\lambda^2} & \lambda(X, X_{ob}) \le \lambda_0 \\ 0 & \lambda(X, X_{ob}) > \lambda_0 \end{cases} \tag{13}$$

The combined force on the target is:

$$F(X) = F_{at}(X) + F_{re}(X) \tag{14}$$

Under this combined force, the target point will move towards the target direction. However, when there are obstacles near the final target, the gravitational force on the target point is relatively small, while the repulsive force given by the obstacles is relatively large, resulting in the inability to reach the predetermined target point[22]; In addition, during the movement process, there may also be situations where the resultant force is zero, resulting in the movement stopping or reciprocating motion at a certain point[23], which will also lead to the inability to reach the predetermined position.

B. Improving the Artificial Potential Field Algorithm

a. Introducing target position function

When there are too many obstacles near the target point, the moving point will be unable to reach the target point due to the strong repulsive force generated by the obstacles, which will cause the drone to move back and forth near the target point, forming an unreachable problem. To ensure that the target can reach the expected point smoothly, the target position function $\left(X - X_g\right)^m$ is added to the repulsion function to change the magnitude of the combined force on the robot near the target point, thereby achieving the expected effect. The improved repulsion function is as follows:

$$U_{re}(X) = \begin{cases} \dfrac{1}{2}\eta\left[\dfrac{1}{\lambda} - \dfrac{1}{\lambda_0}\right]^2 \left(X - X_g\right)^m & \lambda \le \lambda_0 \\ 0 & \lambda > \lambda_0 \end{cases} \tag{15}$$

$$F_{re}(X) = \begin{cases} F_{re1} + F_{re2} & \lambda \le \lambda_0 \\ 0 & \lambda > \lambda_0 \end{cases} \tag{16}$$

Wherein:

$$F_{re1} = \eta\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)\frac{1}{\lambda^2}\left(X - X_g\right)^m \tag{17}$$

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$$F_{re2} = \frac{1}{2}\eta\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)^2 \left(X - X_g\right)^{m-1} \tag{18}$$

Among them, the repulsion direction of F_{re1} is the direction from the obstacle to the robot, and the repulsion direction of F_{re2} is the direction from the robot to the target; Fig 3 show the improved repulsive and gravitational fields, respectively.

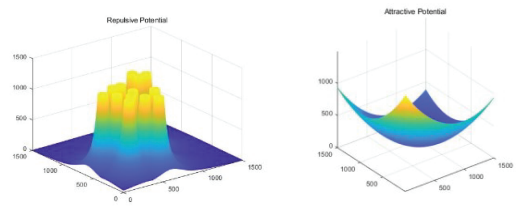

Fig. 3. Improved repulsive and gravitational potential fields

According to equations (17) and (18), we can change the magnitude of repulsion by controlling the magnitude of m.

When $0 < m < 1$,

$$F_{re1} = \eta\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)\frac{\left(X - X_g\right)}{\lambda^2} \tag{19}$$

$$F_{re2} = \frac{1}{2}\eta\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)^2 \frac{1}{\left(X - X_g\right)} \tag{20}$$

F_{re2} is infinitely large, meaning that the direction of force is directed by the robot towards the target.

When $m = 1$,

$$F_{re1} = \eta\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)\frac{\left(X - X_g\right)}{\lambda^2} \tag{19}$$

$$F_{re2} = \frac{1}{2}\eta\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)^2 \tag{20}$$

F_{re1} is infinitely small, and F_{re2} is a fixed value, indicating whether the direction of the force is still pointing towards the target direction.

When $m > 1$,

$$\lim F_{re2} \to 0, \lim F_{re2} \to 0 \tag{17}$$

b. Introducing virtual target points

When a drone is subjected to the same magnitude of gravity and repulsion in the opposite direction, it enters a state of force balance, causing the drone to stop moving [24]. When the drone is in force equilibrium, by assuming a virtual target, the force equilibrium state is disrupted [25], causing the drone to move towards the virtual target, thereby guiding the drone away from the local optimal problem.

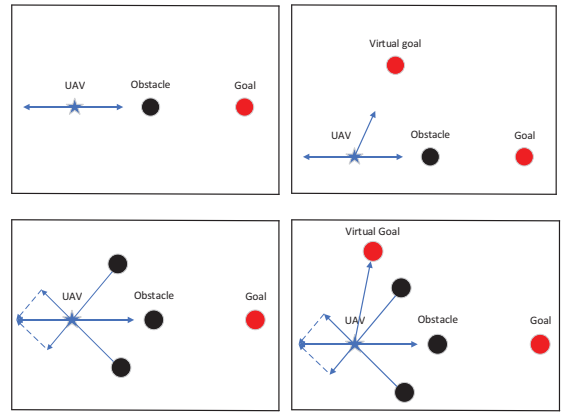

Fig. 4. Comparison of resultant force directions before and after adding virtual targets

Fig 4 shows the changes in the combined force after adding a virtual target point in different obstacle environments. It can be seen that after adding a virtual target point, the original equilibrium state is broken, causing the robot to move towards a new target point. The gravitational expression of the new target point is:

$$F_{vir} = \frac{1}{2}\phi\left|D_n - D_{vir}\right|^2 \tag{24}$$

The overall combined force acting on the drone is:

$$F = F_{rej} + F_{att} + F_{vir} \tag{25}$$

Among them, ϕ is the gain coefficient, which is related to the complexity of the obstacles in the environment where the drone is located. When the drone reaches the set virtual target point, the gravitational value becomes zero.

IV. Simulation and Results

This simulation experiment uses the grid method to model the environment [26], as shown in the following fig 5. Green and red represent the starting and ending points of the robot, while black represents the obstacles set up. The robot will use the improved algorithm in this article to plan its path and cross the obstacles to reach the target point.

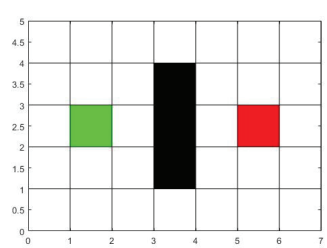

Fig. 5. Environmental Information Model

The A * algorithm can plan a globally optimal path in the overall environment, while the path planned by the artificial

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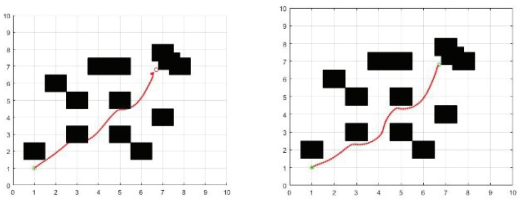
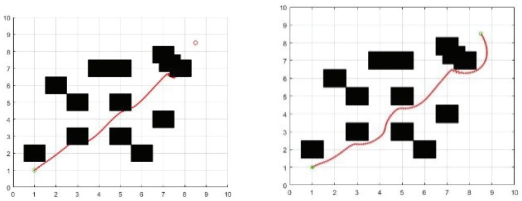
potential field method in the overall environment is often not the optimal path [27]. If only the A * algorithm is used for path planning, there will be a phenomenon of delayed obstacle avoidance, and if the artificial potential field method is used alone, there will be a problem of unreachable targets or local minima, making it impossible to reach the target point [28]. In this article, the improved A * algorithm is first used to plan the global optimal path in a set environment. On the planned path, the improved artificial potential field method is used to assume the turning point of the path as the target point, and the path is segmented and optimized to make the path smoother and obstacle avoidance more timely.



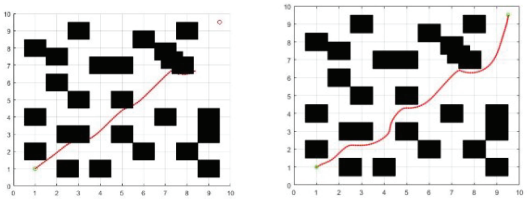
(a) Before improvement (b) After improvement

Fig. 6. The situation where the target point is very close to the obstacle



(a) Before improvement (b) After improvement

Fig. 7. The situation where the target point is located behind an obstacle



(a) Before improvement (b) After improvement

Fig. 8. Path planning in complex environments

From the comparison results in Fig 6, it can be seen that when the target point is very close to the obstacle, the improved algorithm can still plan a suitable path to reach the target point. However, the original algorithm may encounter local minima and cannot reach the target point, and there may be situations where it touches the obstacle during the path planning process; As shown in Fig 7, when the target point is located behind an obstacle, the improved algorithm can bypass the obstacle and reach the target point on a relatively smooth trajectory. However, the original algorithm will terminate its motion due to the strong repulsive force of the obstacle, and ultimately cannot reach the target point; As shown in Fig 8, when in complex environments, compared to the unimproved algorithm, the improved algorithm can plan a smoother path to reach the target point without colliding with obstacles. The unimproved algorithm cannot reach the target point, and the planned path may also collide with obstacles.

V. CONCLUSION

This text described improvements made to the traditional A* algorithm and the traditional Artificial Potential Field (APF) algorithm due to their existing shortcomings. By incorporating a target position function and introducing virtual target points, the issue of drones being unable to reach their destination was resolved; a method of path planning in segments was employed to make the overall path smoother. Additionally, the two improved algorithms were integrated, and through simulation, the following conclusions were drawn:

- By modifying the cost function, the overall path was made shorter, enhancing the efficiency of the algorithm;
- The robot's ability to avoid obstacles in a timely manner was improved, and the path planning was made more reasonable through the secondary optimization of the APF;
- The experimental results showed that the integrated algorithm's planned path was safer, smoother, and more in line with the actual motion trajectory of the robot.

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